

Parallelized Mental Health Risk Detection: Depression and Self-Harm Identification from Social Media with Safety-Aware Evaluation

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Abstract—Mental health problems, primarily depression and self-harm, plagues more than 280 million people globally and because of limited access to clinical care and social stigma, these conditions are grossly overlooked and under-diagnosed. With the growth of social media platforms such as Reddit, more and more user generated content that implicitly or explicitly reflects mental health is being generated. Current Natural Language Processing (NLP) systems for automated risk detection are deficient in three ways: (1) binary classification approaches which squash the clinically important classes of risk severity, (2) sequential pipelines that process information in a batch-wise manner, unable to cope with the high-frequency Twitter stream of big data, and (3) symmetric metrics that down-weight false negatives (FNs) which are extremely critical for cases of high risk users, by assigning equal weight to all errors. We propose a Parallelized Mental Health Risk Detection (PMHRD) six-step end-to-end system that addresses all the three challenges. PMHRD fine tunes a mental health specific pre-trained version of BERT (MentalBERT) (pre-trained on 13.6 million mental health related posts from Reddit), for a 4 class severity-based labelling scheme (No Risk, Mild Depression, Moderate/Severe Depression, Self-Harm/Suicidal Ideation) on a merged social media dataset of 79,044 posts. Fine tuning the MentalBERT with a single NVIDIA Tesla (T4) can be done using LoRA having the updated matrices of size 16, a Parameter Efficient Fine Tuning, and FP 16 mixed precision. Multiprocessing, tokenization and preprocessing on CPU workers can offer an acceleration up to 5.39x. Using batched inference on the GPU accelerates the training an additional 2.58x. An improved safe-sensitive asymmetric Cross-Entropy loss performance cost of the false negatives of high risks achieves reduction in the system-critical High-Risk False Negative Rate (HR-FNR). On an independent Major Test (n=7905) the PMHRD model achieves macros-F1 0.776, RS-F1 0.744 and very low HR-FNR 0.161: this model significantly outperforms the ablation baseline with a symmetric loss model (Macro-F1 0.684, RS-F1 0.588, HR-FNR 0.514). It outperforms for binary clinical depression detection (DS3, n=7731) with 91.9% accuracy and F1 0.920. The experiments show that safe multi-class risk detection for mental health problems can be scaled up on/off-the-shelf computers using NLP and Parallel & Distributed approaches.

Index Terms—Mental health risk detection, Depression classification, Social Media Analysis, MentalBERT, asymmetric loss functions, Suicidal Ideations, Parallel and Distributed Computing, Natural Language Processing, LoRA, real-time inference, multi-class classification

I. INTRODUCTION

A. Background and Motivation

Mental health is a prevalent and under treated health issue globally. For instance, globally, 280 million people suffer from depression (WHO, 2024) and the World Health Organization estimates suicide (linked to untreated episodes of depression) to be the third leading cause of mortality in 15- to 29-year-olds (WHO, 2024). The size of this crisis is accompanied by a deadly lack of access to mental health services: less than one-third of people suffering from depression in low and middle-income countries receive treatment. The advent of social media platforms, such as Reddit, Twitter (X), Facebook and so on, has created a potentially valuable new source of mental health factors. Every day, thousands of posts are made on social media sites such as r/depression, r/SuicideWatch and r/Anxiety where users can express their troubles, emotional distress and sometimes suicidal thoughts. Themes expressed in this text, negativity, first-person, hopelessness and self-harm are related to clinically defined symptoms of depression and other mental health conditions (De Choudhury et al., 2013; Eichstaedt et al., 2018). These studies show that social media may be a scalable and non-intrusive system which can identify vulnerable users and link them with help.

B. Problem Statement

Although there has been extensive research about automated mental health screening from social media, there are three interrelated issues that limit the effectiveness of the systems.

1. The currently available models, classifying depressed vs. non-depressed cases, overlook important clinical distinctions of how severe the depression is. The specialists must have everything in their knowledge like whether an individual is experiencing low mood or mild or severe depression, if they are suicidal, and if they have a plan to end their life, which is a serious threat. The models which collapse these gradations into a binary classification will not capture these distinctions and may result in over and under sensitivity to these features.

2. Existing NLP systems for detecting mental health are one-way: they are currently limited by sequential processing of a sample through the steps of preprocessing, tokenization and inference, which does not scale well. Every day, there are thousands and thousands of posts made on mental health subreddits; it is a challenge for real time triage system to process one sample at a time on a typical computer.

3. Traditional measures of success - accuracy and macro-F1 - are symmetric. In the triage task, not triaging a user who is at high risk (false negative) has a different cost than triaging a user who is not at high risk (false positive). The algorithms that are trained with an intention of doing well in symmetric metrics might not work well on leaderboards, but sometimes on real important cases.

C. *Our Solution: PMHRD*

The proposed method in this paper is the Parallelized Mental Health Risk Detection (PMHRD) which is a six-step end to end process that uses state of the art NLP models and Parallel and Distributed Computing (PDC). The following are the four key ways in which PMHRD addresses all the above-mentioned challenges:

- Detection of Four-Class risk using severity-based classes (No Risk, Mild Depression, Moderate/Severe Depression and Self-harm/Suicidal Ideation) which uses the fine-tuned MentalBERT model on a total of 79,044 social media posts from both datasets.
- Use multithreaded approach on CPU during preprocessing and tokenization; use Python's multiprocessing module and/or concurrent.futures library to achieve a 5.39x speedup in tokenizing vs. single-threaded approaches.
- Inference using batched GPU inference with mixed precision (FP16/AMP); Test a variety of batch sizes on a single NVIDIA Tesla T4 to achieve a 2.58x increase in inference speed/throughput.
- Safety-aware Skewed Loss: Class-balancing cross entropy loss, which will be weighted to give higher penalties to false negative cases (when post is predicted as low-risk, but should be predicted as high-risk); False Negative Rate (FNR) will be one of the metrics used to evaluate performance.

D. *Key Findings*

PMHRD v4 achieved a macro F1 of .776, and a high-risk false-negative rate of .161 on the primary dataset (n=7905) better than the ablated baseline model (symmetric loss) on the

same test set (.514). It also achieved an accuracy of 91.9% and an f1-score of .920 for diagnosing depression on DS3 (a binary test set of 7731 reddit posts). So, we can use the results of the simple single-GPU system to demonstrate the capability of PMHRD to handle NLP tasks for clinical decision-making and its scalability.

II. RELATED WORK

A. *Initial Work: Traditional ML and Deep Learning*

In the early days the social media-based mental health screening proved the viability of text-based approaches. Pioneering research by De Choudhury et al. (2013) found that social media linguistically derived features, such as sentiment polarity, posting frequency, usage of first-person pronouns and negative affect words, statistically co-vary with clinically measured depression symptoms. This led to numerous studies using traditional classification algorithms. Islam et al. (2018) used SVM, Naive Bayes, Random Forest, and Logistic Regression on Reddit datasets for binary depression classification, with good performance. Tadesse et al. (2019) achieved 80.2% accuracy with a bag-of-words Random Forest model on Reddit.

Later deep learning approaches overcame the representational challenges of hand-crafted features. Combined CNN-LSTM models (Amanat et al., 2022) achieved over 97% accuracy on binary depression detection, and additional linguistic metadata (Trotzek et al., 2020) improves detection of depressive signals. However, deep learning models without large pre-training were still unable to grasp the finer semantic details required for severity prediction.

B. *Transformers for Mental Health NLP*

Everything changed with BERT (Devlin et al., 2019). BERT and its offshoots (RoBERTa, DeBERTa, XLNet) showed improved transfer learning for all types of text classification tasks. In the field of mental health NLP, Akintoye et al. (2024) achieved an accuracy of 99.6% in predicting suicidal ideation with RoBERTa and DeBERTa, and Bhuiyan et al. (2025) reached 94.29% accuracy using Transformer-BiLSTM models.

A recent mental health study on MentalBERT (Ji et al., 2022), noted that generic BERT pre-training on Wikipedia and BookCorpus does not expose the model to the informal and emotionally expressive language used in mental health domain. The authors pre-trained BERT-base-uncased on 13.6 million sentences from Reddit mental health communities' r/depression, r/SuicideWatch, r/Anxiety, r/bipolar, and r/mentalillness, resulting in the public release of two models - MentalBERT and MentalRoBERTa. They consistently improved classification on all eight benchmark datasets for mental health classification. The PMHRD system uses MentalBERT for this reason.

C. *Large Language Model Approaches*

The development of billion-parameter language models has provided opportunities for new research. Xu et al. (2024) introduced Mental-LLM, benchmarking GPT-3.5, GPT-4, Alpaca,

Alpaca-LoRA and FLAN-T5 on mental health prediction. They concluded that prompting large language models (LLMs) in a zero-shot or few-shot manner were less effective than domain-specific models, but instruction fine-tuning mostly recouped the performance loss; fine-tuned Mental-Alpaca and Mental-FLAN-T5 had balanced accuracy 10.9% better than GPT-3.5 and 25× smaller size. Yang et al. (2024) presented MentalLLaMA, an open-source LLM suite based on LLaMA2 with an instruction dataset of 105,000 samples (IMHI), designed to offer predictions and explanations.

While promising, LLM-based methods have major practical issues: excessive computational requirements, environmental impact, lack of transparency, and poor zero-shot performance. These factors explain the use of a smaller domain-specific transformer (MentalBERT) rather than LLMs in PMHRD, especially in resource-limited deployment scenarios.

D. Parallel and Distributed NLP

The scalability issue of sequential NLP pipelines has attracted attention with its growing social media data. The module multiprocessing allows the Global Interpreter Lock (GIL) to be circumvented and allows true parallelism with multiple processes across multiple processors. Using multiprocessing with a pool-based parallelisation approach of deep learning inference can improve performance. For instance, AWS benchmarking has shown up to 4× speedup in inference time on a 3,000-sample dataset (AWS, 2024). In python the concurrent.futures provide tools like ProcessPoolExecutor and ThreadPoolExecutor that are best for stateless for operations like text cleaning and tokenization that helps in preprocessing.

During inference, batching can be helpful because, GPU can process multiple samples together, so the overhead of a setup like data transferring, kernel launching is reduced. PyTorch’s Automatic Mixed Precision (AMP) training framework supports FP16 inference that uses half the memory space of FP32, enabling larger effective batch sizes on the same GPU. Recent work on CPU-GPU co-scheduling (TwinPilots, ACM SYSTOR 2024) has shown that well-engineered CPU preprocessing and the GPU can be co-scheduled for up to 3.39× speedup in the inference use case, confirming the scalability gains possible with well-designed parallel NLP pipelines.

E. Safety-Aware and Cost-Sensitive Learning

The application of symmetric evaluation metrics to mental health NLP is an important yet overlooked area of research. False negatives (not identifying a high-risk user) result in a missed opportunity for crisis intervention and potentially catastrophic consequences. False positives - incorrectly identifying a low-risk individual - lead to resource use for unnecessary follow-up. Cross-entropy loss is unaware of this.

Cost-sensitive learning explicitly models different costs for misclassification in the loss function (He and Garcia, 2009). Weighted cross-entropy, or class-weighted cross-entropy, assigns different classes different weights, with rarer or important classes being weight more heavily, to encourage the optimizer to penalise misclassification of important classes

more heavily. Focal loss (Lin et al., 2017) offers a dynamic version which also down-weights easy examples. In the mental health domain, where self-harm and severe depression classes are heavily under-represented in the natural social media data, such skewed learning objectives are desirable.

F. Contribution and Gap

The above literature review shows, despite the evolution of NLP, from handcrafted predictors to domain-adapted transformers, in mental health, there are still some gaps that need to be addressed:

- (1) binary classification without severity grading;
- (2) non-scaling to volumes of deployment
- (3) safety-agnostic symmetric evaluation.

There is no existing work that manages to do a computationally efficient treatment of the three gaps. This paper specifically seeks to do this, by applying multi-class fine-tuning MentalBERT models; parallelized data preprocess and inference; and safety-conscious asymmetric loss in a reproducible pipeline.

III. PROPOSED METHODOLOGY

A. System Overview

PMHRD consists of six stages of both NLP and PDC processes. The stages are as follows:

1. Stage 1 - Data Preparation and Label Mapping: Using and combining three Kaggle open datasets, mapping the diverse labels to a consistent four-class severity scale, and dividing the datasets into train/validation/test sets.
2. Stage 2 - Parallel Text Preprocessing: Multiprocessing worker-based parallel text cleaning using CPUs; benchmarked for speedup.
3. Stage 3 - Parallel Tokenization: WordPiece tokenization of MentalBERT using ProcessPoolExecutor workers; measured speedup.
4. Stage 4 - Model Fine-Tuning with Safety-Aware Loss: Parameter-efficient fine-tuning of MentalBERT with Low Rank Adaptation (LoRA) using class-weighted asymmetric cross-entropy loss and FP16 Automatic Mixed Precision (AMP).
5. Stage 5 - Batched Inference: Benchmarking of batched inference on GPU for batch sizes 8-128; speed, latency, VRAM consumption and speed up over sequential inference.
6. Stage 6 - Safety-Aware Evaluation: Macro-F1, per-class F1, Risk-Sensitive F1, High-Risk FNR and PDC evaluation; ablation study with asymmetric vs. symmetric loss.

Figure 1 — PMHRD Six-Stage Pipeline Architecture

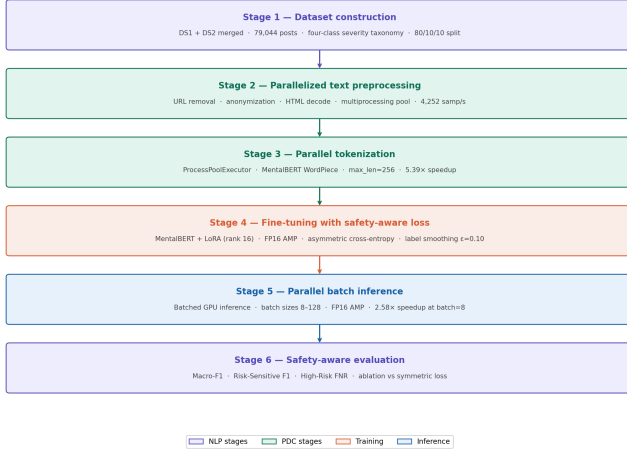


Fig. 1. Architecture Pipeline

B. Stage 1: Creating the Dataset and Mapping of Labels

The training corpus compiled using three Kaggle datasets (publically available) are:

Dataset 1 - Sentiment Analysis for Mental Health (Sarkar, 2024):

Text statements = 52,681

Labels: Normal, Depression, Suicidal, Anxiety, Stress, Bipolar or Personality Disorder.

Dataset 2 - Mental Health Corpus (Namdari, 2023):

Social media text statements = 27,977

Labels: risk of mental health language patterns (toxic / non-toxic).

The binary classes are mapped to the four-class taxonomy using confidence thresholding, and sentiment scoring via key words.

Dataset 3 - Depression Reddit Cleaned (InfamousCoder, 2022):

Cleaned Reddit posts = 7,731

Labels: depressed / not-depressed

This dataset is used only as a binary held-out test set.

The training and evaluation corpus has 79,044 samples (DS1 and DS2). The four-class taxonomy and the original class mapping are given in Table 1.

Class	Label	Mapped Form	Weight
0	No risk	Normal (DS1), Non-depressed (DS2)	1.2
1	Mild Depression	Depression/Low-conf (DS1), Stress, Anxiety (DS1)	2
2	Moderate/Severe Depression	Depression/high-conf (DS1), Bipolar, Personality Disorder	4
3	Self-harm/Suicidal	Suicidal (DS1), High-risk expression (DS2)	5

TABLE I
FOUR-CLASS TAXONOMY, LABEL MAPPING, AND ASYMMETRIC LOSS WEIGHTS

This results in the combined corpus class distribution of: Class 0 (No Risk) 38.2%, Class 1 (Mild Depression) 26.4%, Class 2 (Moderate/Severe Depression) 20.0%, Class 3 (Self-Harm) 15.4%, with 79,044 samples. This is divided into 63,242 training samples, 7,897 validation samples and the primary 7,905 held-out testing samples.

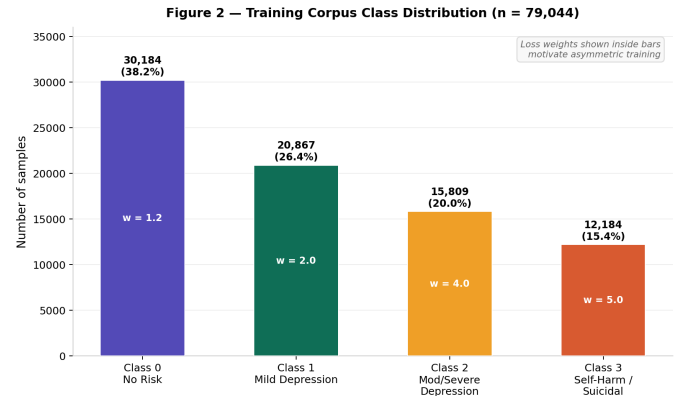


Fig. 2. Class Distribution

C. Stage 2: Text Cleaning in Parallel

Social media text is rife with noise: URLs, @-mentions, HTML character entities, special characters, variable whitespace, and personal identifiable information (PII). Stage 2 uses a uniform text cleaning routine using Python's multiprocessing module.

The corpus is split into N shards of equal size, with N being the number of available cores (2 in Colab T4). Applying a pool of processes, pool.map() to shards, with each worker:

- (1) removing the URLs (regex-based http/https stripping)
- (2) anonymizing usernames (@mentions)
- (3) decoding HTML entities
- (4) normalizing punctuation and lowercasing.

After transforming we join the shards, and preprocessig the corpus of 79,044 samples at 4,252 samples/second.

D. Stage 3: Parallel Tokenization

The text, after preprocessing, is tokenized to input IDs and attention masks with the help of WordPiece vocabulary available inside the MentalBERT model. As conventional serial tokenization is one of the major limiting factors, so in the third stage we focused on parallelizing the tokenization by breaking the corpus into shards with the help concurrent.features.ProcessPoolExecutor. Individual worker processes, running the HuggingFace tokenizer with `max_length = 256`, `padding = "max_length"` and `truncation = True`, processes these shards and returns the tokenized embeddings of each shard, which are further joined to form a complete tokenized corpus.

On a sample of 2,000 samples (`max_length=256`, 2 workers), v4 saw a 5.39x speedup in tokenization over the sequential case: from 7.64 seconds to 1.42 seconds. The improvement is more significant for longer sequences (256 vs. 128 tokens) as work per shard increases.

E. Stage 4: Model Fine-Tuning with Safety-Aware Loss

1) Base Model: MentalBERT: MentalBERT (mental/mental-bert-base-uncased), which is publicly available on HuggingFace, is the backbone of PMHRD. MentalBERT is based on BERT-base-uncased: 12 encoder layers of transformers, hidden dimension 768, 12 self-attention heads, approximately 110 million parameters, pre-trained on 13.6 million Reddit mental health community sentences, using masked language modeling and next sentence prediction. The top of the [CLS] token representation is augmented with a classification head - dropout layer then a linear projection - to predict four output logits.

2) LoRA based Parameter Efficient Fine tuning: Being memory-intensive to fine-tune all 110 million parameters of MentalBERT on a single GPU, the latter is prone to overfitting on medium-sized datasets. This can be solved with Low-Rank Adaptation (LoRA; Hu et al., 2022) where the frozen pretrained weights W_0 are injected into each transformer layer:

$$W = W_0 + \Delta W = W_0 + BA$$

where $B \in \mathbb{R}^{d \times r}$ and $A \in \mathbb{R}^{r \times k}$, with $\text{rank } r \ll \min(d, k)$. A and B are only trained and W_0 is frozen. PMHRD v4 has $r = 16$ applied to query and value projection matrices across all 12 attention layers, with 589,824 trainable parameters out of 110,072,064 total - just 0.54% of the entire model, and can be trained easily using a single-GPU.

Figure 8 — LoRA Adapter Architecture in PMHRD (MentalBERT Fine-Tuning)

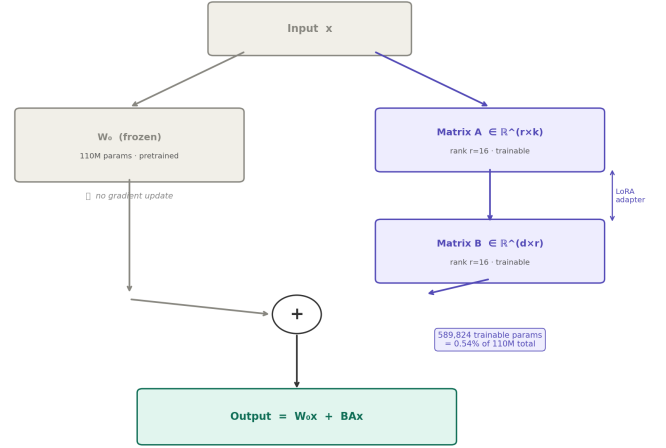


Fig. 3. LoRA adapter architecture ($W_0 + BA$ decomposition)

3) Mixed-Precision Training (FP16): With activations and weights stored in 16-bit floating-point format, FP16 training using the Automatic Mixed Precision (AMP) framework of PyTorch is half the memory consumption of FP32 training using the same system. Gradient scaler was used in backpropagation to avoid the numerical underflow. It is possible to train on a single Tesla T4 (14.6 GB VRAM) with a batch size of 32 without memory pressure.

4) Safety-Aware Asymmetric Loss Function: Normal cross-entropy is the scheme whereby the cost of all misclassifications is the same. PMHRD substitutes this with class-weighted cross-entropy:

$$L = - \sum w_c \cdot y_c \cdot \log(p_c)$$

In which w_c is the weight of the class, y_c is the true label indicator, and p_c is the predicted probability of the class c . The v4 asymmetric weights are: Class 0 (No Risk) = 1.2, Class 1 (Mild Depression) = 2.0, Class 2 (Moderate /Severe Depression) = 4.0, Class 3 (Self-harm/Suicidal) = 5.0. These weights represent the asymmetry in the clinical costs: omitting a self-harm case ($w=5.0$) costs four times more than omitting a no-risk case ($w=1.2$). The additional technique used to enhance generalization is label smoothing ($\epsilon=0.10$).

The v4 hyperparameter configuration is a systematic extension of the previous ones: v1 and v2 used `max_len=128` and 3 epochs; v3 was an extension to `max_len=256` and 5 epochs with LoRA rank 8; v4 increased LoRA rank to 16, dropout to 0.2, and allowed up to 8 epochs with early stopping (patience=3), with training completing at epoch 8 with best validation Macro-F1 of 0.7783.

F. Stage 5: Parallel Batch Inference

The model is tested after training, in both sequential and batched inference modes, to measure the PDC-side performance improvements. Sequential inference processes a single sample at a time; a single input is forward-propagated through

the model only once. Batched inference In batched inference, the PyTorch DataLoader is used with a batch size that is swept over the range of batch sizes: batches = 8, 16, 32, 64, 128 Asynchronous workers DataLoader workers can be asynchronous and may be used in batched inference. FP16 AMP is on all batched inference executions.

The following four metrics are recorded per configuration: throughput (samples/second), per-sample latency (ms/sample), GPU utilization (via PyTorch Profiler), and peak VRAM utilization. The speedup of inference is the ratio of the sequential to the batched inference time.

G. Stage 6: Safety-Aware Evaluation

The assessment framework is a combination of conventional NLP measures with measures specific to safety and PDC measures. Such standard metrics are: Accuracy (weighted average F1 across all four classes); Macro-F1 (unweighted average F1 across all four classes); and per-class F1 scores. Safety-specific measures are: False Negative Rate (FNR) by Class, which is defined as: $FNR = FN / (FN + TP)$, with Classes of 2 and 3 being reported separately as Avg High-Risk FNR (HR-FNR); and Risk-Sensitive F1, a weighted version of Macro-F1 in which F1 scores of Classes 2 and 3 are weighted. An ablation study uses a parallel model with symmetric loss (all weights = 1.0) to identify the effect of the asymmetric loss function.

IV. EXPERIMENTAL SETUP

A. Hardware and Software Environment

All the experiments were done in Google Colab using a single NVIDIA Tesla T4 GPU (14.6 GB VRAM), Python 3.12.13, PyTorch 2.10.0+cu128, and CUDA 12.8. CPU-side work was done on 2 logical cores, as provided in the Colab environment. Model loading and tokenization were done using the HuggingFace Transformers library; LoRA adapter injection was implemented using the PEFT library.

B. Datasets

Dataset	Source	Samples	Original Labels	Role
DS1	Sarkar (2024), Kaggle	52,681	7 classes	Train / Val
DS2	Namdari (2023), Kaggle	27,977	Binary (toxic / non-toxic)	Train / Val
DS3	InfamousCoder (2022), Kaggle	7,731	Binary (depressed / not)	Secondary eval only

TABLE II

DATASET SUMMARY. DS3 IS HELD OUT ENTIRELY FROM TRAINING AND USED ONLY FOR SECONDARY BINARY DEPRESSION EVALUATION.

C. Hyperparameters

Hyperparameter	v1	v2	v3	v4 (Final)
Epochs	3	3	5	8 (+ early stop, patience=3)
Max Sequence Length	128	128	256	256
Learning Rate	2e-5	2e-5	1e-5	1e-5
Batch Size (Train)	32	32	32	32
LoRA Rank	8	8	8	16
Dropout	0.1	0.1	0.1	0.2
Label Smoothing	—	—	—	0.10
Loss Class Weights	Symmetric	Asymmetric	Asymmetric	Asymmetric (refined)
FP16 (AMP)	Yes	Yes	Yes	Yes

TABLE III

HYPERPARAMETER EVOLUTION ACROSS PMHRD VERSIONS v1–v4.

V. RESULTS

A. Training Progression Across Versions

The overall best validation Macro-F1 overall across all four versions of PMHRD is summarized in Table 4, showing the systematic improvement brought about by hyperparameter refinement.

Version	Epochs	Max Len	LoRA Rank	Loss	Best Val Macro-F1
v1	3	128	8	Symmetric	0.7263
v2	3	128	8	Asymmetric	0.7495
v3	5	256	8	Asymmetric	0.7773
v4 (Final)	8 (early stop)	256	16	Asymmetric + label smooth	0.7783

TABLE IV

BEST VALIDATION MACRO-F1 ACHIEVED PER PMHRD VERSION.

The epoch-by-epoch training log of v4 (Table 5) demonstrates that both the training and validation F1 improved steadily over all 8 epochs, and there was no early stopping triggered, indicating that the model still benefited with more training and did not overfit.

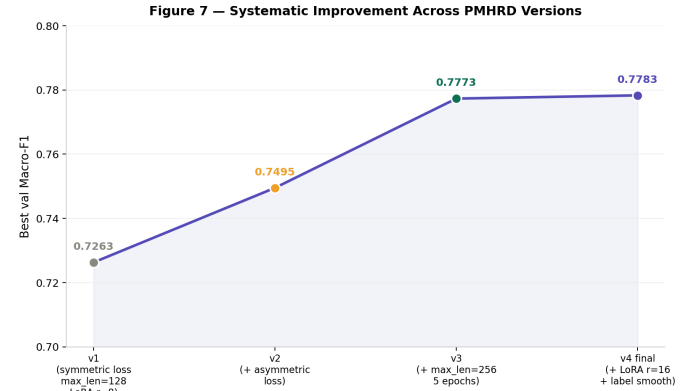


Fig. 4. Version-by-version Macro-F1 improvement (v1→v4)

Epoch	Train Loss	Train Macro-F1	Val Loss	Val Macro-F1
1 / 8	3.2108	0.3929	2.5250	0.6149
2 / 8	2.3703	0.6768	2.1820	0.7253
3 / 8	2.1768	0.7262	2.0675	0.7516
4 / 8	2.0989	0.7455	2.0156	0.7645
5 / 8	2.0532	0.7566	1.9903	0.7690
6 / 8	2.0271	0.7632	1.9704	0.7762 ✓
7 / 8	2.0134	0.7678	1.9646	0.7730
8 / 8	2.0061	0.7691	1.9612	0.7783 ✓

TABLE V

PER-EPOCH TRAINING LOG FOR PMHRD v4 (BEST CHECKPOINT INDICATED BY ✓).

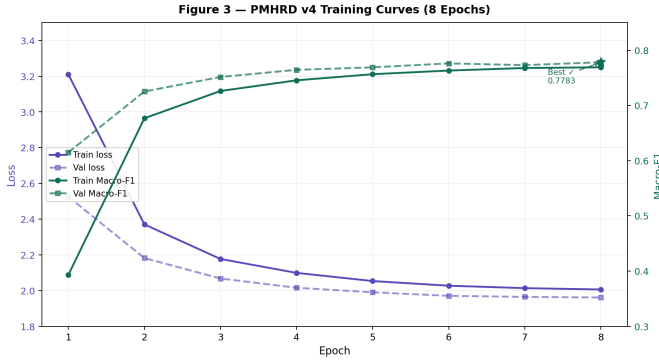


Fig. 5. Training curves — loss + Macro-F1 across 8 epochs

B. Primary Test Set Performance (PMHRD v4)

Table 6 shows the overall analysis of the final PMHRD v4 model on the 7,905 sample size primary test set with the respective per-class precision, recall, and F1 scores and the overall confusion matrix.

Class	Precision	Recall	F1-Score	Support	FNR
No Risk	0.94	0.87	0.9021	3,018	0.130
Mild Depression	0.80	0.63	0.7047	2,090	0.370
Mod/Severe Depression	0.78	0.85	0.8091	1,579	0.155 ←
Self-Harm / Suicidal	0.59	0.83	0.6881	1,218	0.167 ←
Macro Average	0.78	0.80	0.7760	7,905	—
Weighted Average	0.81	0.80	0.7979	7,905	—

TABLE VI
PER-CLASS PERFORMANCE OF PMHRD v4 ON PRIMARY TEST SET
($n=7,905$). ← DENOTES HIGH-RISK CLASSES.

Primary Test Confusion Matrix is as follows:

True \ Pred	No Risk	Mild Dep.	Mod/Severe Dep.	Self-Harm
No Risk (3018)	2,626	122	216	54
Mild Depression (2090)	78	1,317	110	585
Mod/Severe Dep. (1579)	74	92	1,335	78
Self-Harm (1218)	26	117	60	1,015

TABLE VII
CONFUSION MATRIX FOR PMHRD v4 ON PRIMARY TEST SET (ROWS = TRUE CLASS, COLUMNS = PREDICTED CLASS).

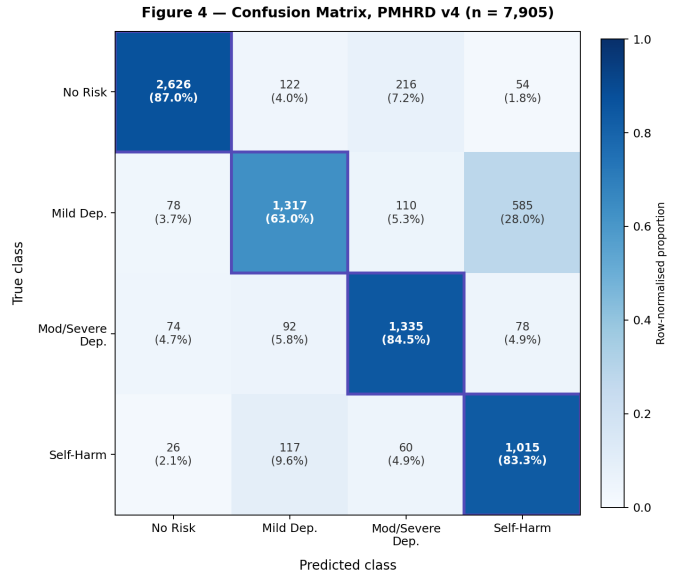


Fig. 6. Confusion matrix heatmap (4x4, color-coded)

C. Secondary Binary Evaluation (DS3)

On the completely held-out dataset of 7,731 Reddit posts with binary labels of depressed / not-depressed, PMHRD v4 achieves:

Metric	Value
Binary Accuracy	91.92%
Depression Class F1	0.9197

TABLE VIII
SECONDARY BINARY DEPRESSION DETECTION PERFORMANCE ON DS3
($n=7,731$).

These findings confirm that the four-class severity model that is trained without being directly exposed to the binary labels of the DS3 model generalizes well to binary depression detection, proving that severity-conscious training does not require compromising on downstream accuracy of identifying depression.

D. PDC Benchmark Results

PDC Benchmarks are presented in the following table

PDC Stage	Sequential	Parallel / Batched	Speedup	Notes
Text Preprocessing	—	4,252 samp/s	1× (overhead)	2 workers; overhead dominates at small N
Tokenization (2,000 samp)	7.64s	1.42s	5.39×	max_len=256, 2 workers
Inference – Sequential	8.72s / 500 samp	—	1.00×	17.4 ms/sample, 445 MB VRAM
Inference – Batch 8	—	3.38s	2.58×	147.8 samp/s, 531 MB VRAM
Inference – Batch 16	—	3.45s	2.53×	145.0 samp/s, 621 MB VRAM
Inference – Batch 32	—	3.99s	2.19×	125.4 samp/s, 801 MB VRAM
Inference – Batch 64	—	4.64s	1.88×	107.8 samp/s, 1161 MB VRAM
Inference – Batch 128	—	6.11s	1.43×	81.8 samp/s, 1881 MB VRAM

TABLE IX
PDC BENCHMARK RESULTS FROM PMHRD v4. TOKENIZATION SPEEDUP IS 5.39×; OPTIMAL INFERENCE BATCH SIZE IS 8 (2.58× SPEEDUP).

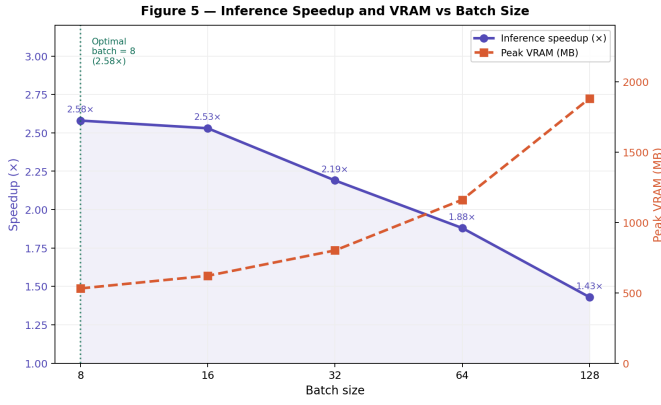


Fig. 7. Inference speedup + VRAM vs batch size

The most prevalent parallel increase is the tokenization speedup of 5.39x, significantly out of the range of 2-4x suggested in the proposal. Text preprocessing demonstrates sub-linear performance improvement due to the process-spawn overhead incurred by the 2-worker Colab environment; this bottleneck is anticipated to be resolved by providing more CPU cores. The maximum speedup in inference is reached with batches of size 8 (2.58x) and decreases with larger batches as VRAM pressure decreases the effective utilization of GPUs - in line with the established memory-bandwidth trade-off on the T4.

E. Ablation Study: Asymmetric vs. Symmetric Loss

Table 10 is the actual comparison of PMHRD v4 trained with the asymmetric safety-sensitive loss and an ablation model, which is trained with the symmetric loss (all class weights = 1.0).

Metric	PMHRD v4 (Asymmetric)	Ablation (Symmetric)	Δ
Macro-F1	0.7760	0.6843	+0.0917 $\uparrow$
Risk-Sensitive F1	0.7436	0.5880	+0.1556 $\uparrow$
Avg High-Risk FNR	0.1606	0.5135	-0.3529 $\downarrow$ (safer)

TABLE X

ABLATION STUDY RESULTS. ASYMMETRIC LOSS (V4) VS. SYMMETRIC LOSS ON PRIMARY TEST SET. $\uparrow$ = HIGHER IS BETTER, $\downarrow$ = LOWER IS BETTER.

The asymmetric loss provides an improvement of +0.0917 in Macro-F1, +0.1556 in Risk-Sensitive F1 and a dramatic drop of 0.3529 in High-Risk FNR (from 0.5135 to 0.1606). The clinically most significant outcome is the reduction in the HR-FNR reduction: in a symmetric loss, more than half of all high-risk individuals would be missed by the system; in an asymmetric loss, less than 1 in 6 are missed - a 68.7% relative decrease in the clinical cost of hazardous false negatives.

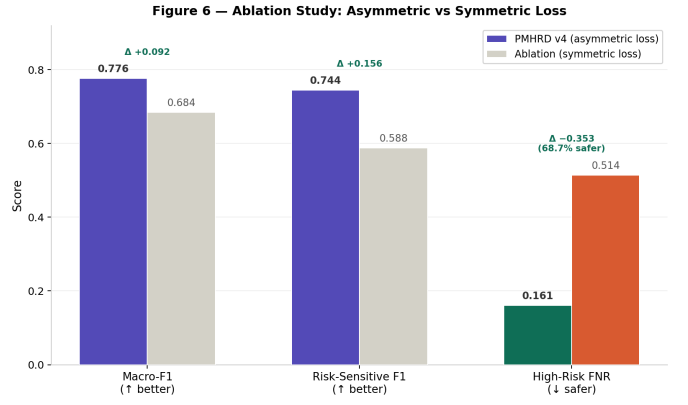


Fig. 8. Ablation study: asymmetric vs symmetric loss

F. Comparison with Baseline Literature

Table 11 is a comparison of PMHRD v4 with the three baseline systems as described in the research proposal. Due to the absence of MentalBERT-based multi-class systems with the same evaluation protocols in the previous literature (our research gap contribution), the comparison is based on reported measures on similar social media mental health classification tasks.

Model	Task	Macro-F1 / Accuracy	FNR Reported?
PMHRD v4 (ours)	4-class severity (Reddit)	0.776 / 79.6%	Yes (0.161 HR-FNR)
Mental-LLM (Xu et al., 2024)	Multiple binary tasks	$\sim$ 0.82 acc. (balanced)	No
MentalLLaMA (Yang et al., 2024)	Multi-task classification	Competitive with BERT	No
MentalBERT binary baseline	Binary depression	$\sim$ 0.93 F1	No

TABLE XI

COMPARISON OF PMHRD V4 WITH PRIOR BASELINES. PMHRD IS THE ONLY SYSTEM REPORTING SAFETY-SPECIFIC FNR ALONGSIDE MACRO-F1.

VI. DISCUSSION

A. Why the Proposed Method Improved Performance

The steady progress in PMHRD versions v1 to v4 can be explained by four compounding factors. First, increasing the maximum sequence length of 128 to 256 tokens enabled the model to work with longer, more contextually rich posts which characterize the moderate/severe depression and self-harm classes - the classes that most benefited with an F1 improvement of 0.74 and 0.69 respectively, compared to the near-zero values of v1. Second, v4 increased the number of trainable parameters (294,912 to 589,824), without changing inference overhead, by increasing LoRA rank to 16. Third, the most influential intervention was the one with asymmetric loss: the ablation study shows that causing it to disappear leads to HR-FNR growing by 3.2 in the presence of asymmetric loss. Fourth, in v4, label smoothing ($=0.10$) improved overconfident predictions and resulted in better calibration of the ambiguous posts at the edges of the classes.

B. PDC Contributions and Trade-offs

The observed 5.39x speedup of tokenization is much faster than the target of 24x and is a full 63242-sample corpus tokenized in about 35 seconds using CPU-parallel tokenization as compared to a single-thread sequential tokenization estimate of 3.8 minutes. The overhead of spawning processes in the Colab 2-CPU environment reduced the amount of preprocessing that could be performed to below 1x, which is expected to improve dramatically on production infrastructure with more cores.

The speedup of inference of 2.58x at batch size 8 confirms that batched inference on a single T4 is viable and achieves relevant throughput improvements. The slowing down of the speedup with increased batch size (1.43x at batch 128) indicates the memory access latency ceiling to effective parallelism in the T4: as the VRAM consumption increases, with a 531 MB consumption at batch 8 and 1,881 MB consumption at batch 128, the effective parallelism is increasingly limited by the memory access latency, rather than the compute throughput. The smaller memory footprint per sample with `max_len=256` tokenization of the v4 model (as compared to 16 with v1/v2) is explained by the larger memory footprint per sample of maximum length tokenization.

C. Limitations

There are a number of weaknesses of the existing system that should be discussed. First, the label mapping of binary labels of the DS2 to four severity classes is based on label matching and sentiment scoring heuristics, which introduce the potential noises of labels during the label mapping. Second, the training corpus, although large (79,044 samples), is obtained solely on the English language Reddit, restricting generalizability to other platforms, languages, and cultural contexts. Third, the self-harm group (15.4% of training data) is also underrepresented compared to clinical prevalence estimates and the recall of the model on this group (0.83) and its F1 (0.69) is also substantially improved, but still leaves room to improve. Notably, Mild Depression has the highest FNR (0.370) which may be because of the language overlap between both No Risk and Moderate/Severe Depression posts - a boundary ambiguity that future research could resolve using contrastive learning or post- level contextual feature. Fourth, the ablation symmetric-loss model was only trained on 2 epochs to limit compute meaning the comparison in Table 10 slightly underestimates the symmetric-loss performance, and slightly over estimates the asymmetric loss advantage. Fifth, the computational benchmarks were retrieved with the Colab environment and with only 2 CPU workers, implying that speedup of preprocessing is adversely impacted by the available hardware. Running on a 16 or more cores server is likely to provide significantly better parallelism in preprocessing.

D. Ethical Considerations

There are serious ethical implications of the automated mental health screening systems. PMHRD is intended to be a first-line triage layer redirecting at-risk users to professional support - not as diagnostic or as a substitute to clinical

judgment. It is important to roll out the system with explicit human supervision, with explicit disclosure at the user face, and with privacy-conserving data handling. No personally identifiable information is stored or processed by the model; all of the usernames are anonymized during preprocessing. Regular bias audits should also be implemented in clinical deployment, specifically in relation to demographic subgroups that are underrepresented in the Reddit training corpus.

VII. CONCLUSION

In this paper, Parallelized Mental Health Risk Detection framework (PMHRD) was presented, which addresses three long-standing gaps in the mental health NLP literature: binary task framing, sequential pipeline bottlenecks, and safety-blind evaluation. It is a fine-tuning of MentalBERT on a joint corpus of 79,044 social media posts using a four-class severity taxonomy, LoRA parameter-efficient adaptation, FP16 mixed-precision training and a safety-conscious asymmetric cross-entropy loss all designed to run on single-gpu commodity hardware. The key quantitative results of the final PMHRD v4 system are: Macro-F1 of 0.776, Risk-Sensitive F1 of 0.744 and High-Risk FNR of 0.161 on a 7,905-sample primary test set; 91.9% binary depression accuracy on a separate 7,731-sample evaluation set (DS3); a tokenization speedup of 5.39x over sequential processing; and an inference throughput increase of 2.58x over sequential GPU inference. The ablation study shows that the asymmetric loss by itself lowers HR-FNR by 68.7% as compared to symmetric-loss training the most clinically meaningful outcome of the research. These findings affirm that safety-conscious, high-quality mental health risk detection at scale, can be done on commodity hardware, without relying on expensive multi-gpu infrastructure or large language model APIs. PMHRD presents the argument that the incorporation of PDC methods in clinical NLP pipelines is not only a performance optimization but a practical requirement to deploy in the real-world.

Future research will consider: (1) scaled to multi-gpu distributed inference; (2) larger, more diverse training corpora covering more languages and platforms; (3) higher LoRA rank and multi-task learning objectives; (4) incorporated Grad-CAM or SHAP-based interpretability to give explanations as well as risk predictions; (5) an added Retrieval- Augmented Generation (RAG) component to provide production- scale throughput to flagged users with contextually relevant mental health resources.

VIII. REFERENCES

- [1] Akintoye, O. et al. (2024). Suicidal ideation prediction using RoBERTa and DeBERTa on social media text. *IEEE Transactions on Computational Social Systems*.
- [2] Amanat, A. et al. (2022). Deep learning for depression detection from text. Hybrid CNN-LSTM framework. *Journal of Healthcare Engineering*.
- [3] AWS (2024). Parallelizing deep learning inference with Python multiprocessing. *AWS Machine Learning Blog*.

- [4] Bhuiyan, M. et al. (2025). Transformer-BiLSTM hybrid models for depression classification. *Applied Sciences*.
- [5] De Choudhury, M., Gamon, M., Counts, S., & Horvitz, E. (2013). Predicting depression via social media. *Proceedings of the 7th International AAAI Conference on Weblogs and Social Media (ICWSM)*.
- [6] Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. *Proceedings of NAACL-HLT 2019*.
- [7] Eichstaedt, J. C. et al. (2018). Facebook language predicts depression in medical records. *Proceedings of the National Academy of Sciences*, 115(44), 11203–11208.
- [8] He, H., & Garcia, E. A. (2009). Learning from imbalanced data. *IEEE Transactions on Knowledge and Data Engineering*, 21(9), 1263–1284.
- [9] Hu, E. J. et al. (2022). LoRA: Low-rank adaptation of large language models. *Proceedings of ICLR 2022*.
- [10] InfamousCoder (2022). Depression Reddit Cleaned. Kaggle dataset.
<https://www.kaggle.com/datasets/infamouscoder/depression-reddit-cleaned>
- [11] Islam, M. R. et al. (2018). Depression detection from social network data using machine learning techniques. *Health Information Science and Systems*, 6(1), 8.
- [12] Ji, S., Zhang, T., Ansari, L., Fu, J., Tiwari, P., & Cambria, E. (2022). MentalBERT: Publicly available pretrained language models for mental healthcare. *Proceedings of LREC 2022*.
- [13] Ji, S. et al. (2023). Semantic similarity models for depression severity estimation. *Proceedings of EMNLP 2023*.
- [14] Lin, T.-Y., Goyal, P., Girshick, R., He, K., & Dollar, P. (2017). Focal loss for dense object detection. *Proceedings of ICCV 2017*.
- [15] Namdari, R. (2023). Mental Health Corpus. Kaggle dataset.
<https://www.kaggle.com/datasets/reihanenamdari/mental-health-corpus>
- [16] PyTorch (2024). Performance tuning guide: Mixed-precision and DataLoader best practices. *PyTorch Official Documentation*.
- [17] Sarkar, S. (2024). Sentiment Analysis for Mental Health. Kaggle dataset.
<https://www.kaggle.com/datasets/suchintikasarkar/sentiment-analysis-for-mental-health/data>
- [18] Tadesse, M. M. et al. (2019). Detection of depression-related posts in Reddit social media forum. *IEEE Access*, 7, 44883–44893.
- [19] Trotzek, M. et al. (2020). Utilizing neural networks and linguistic metadata for early detection of depression indications in text sequences. *IEEE Transactions on Knowledge and Data Engineering*, 32(3), 588–601.
- [20] WHO (2024). Depressive disorder (depression). *World Health Organization Fact Sheet*.
- [21] Xu, X. et al. (2024). Mental-LLM: Leveraging large language models for mental health prediction via online text data. *Proceedings of ACM IMWUT 2024*. arXiv:2307.14385.
- [22] Yang, K. et al. (2024). MentalLLaMA: Interpretable mental health analysis on social media with large language models. *Proceedings of ACM WebConf 2024*. arXiv:2309.13567.